

CP260 Robotic Perception

Final Project Report

Metric-Semantic 3D Reconstruction

1. Abstract

This report presents a complete Metric-Semantic 3D Reconstruction pipeline designed to localize hardware ports on a PC tower's back panel using only posed RGB images. The system employs GroundingDINO — a zero-shot, language-guided object detector — to identify three target entities across 16 camera views: a VGA socket, an Ethernet socket, and a power socket. Two-dimensional detections are lifted to three-dimensional space through multi-view ray triangulation, and Oriented Bounding Boxes (OBBs) are estimated for each entity. The pipeline operates entirely without depth sensors and requires no task-specific training data. Final OBB predictions are exported in a structured JSON format, validated against a provided VGA ground-truth anchor.

2. Introduction

Metric-Semantic Reconstruction is a core problem in robot perception: given a set of images from known viewpoints, can a robot not only reconstruct geometry but also assign meaningful semantic labels to objects — and localize them in metric 3D space with enough precision to act on them?

This project tackles a constrained instance of the problem: localizing small, texture-rich hardware ports (VGA, Ethernet, and power sockets) on the back panel of a PC tower. These are challenging targets because:

- They are small relative to the scene (~1–7 cm across)
- They are partially occluded or foreshortened in many views
- They share similar color and texture with surrounding panel surfaces
- No labeled training data is provided

The approach taken in this project leverages three major components:

1. Zero-shot visual grounding via GroundingDINO to produce 2D detections from natural language prompts
2. Multi-view geometry to triangulate 3D positions from multiple camera rays
3. Depth-based extent estimation to produce full OBB specifications

A key design decision is to use the provided ground-truth VGA socket OBB as a scene anchor — calibrating rotation estimates for all sockets (which share the same panel face) and validating the 3D reprojection accuracy of the pipeline.

2.1 Literature Review — State of the Art

The problem of metric-semantic 3D reconstruction sits at the intersection of neural rendering, open-vocabulary detection, and multi-view geometry. Below is a brief survey of the most relevant SOTA methods:

- **3D Gaussian Splatting (3DGS) — Kerbl et al. (2023):** The current leading method for real-time novel view synthesis. Rather than an implicit volumetric representation, 3DGS represents a scene as a set of anisotropic 3D Gaussians with learnable colour, opacity, and covariance. Rasterization-based rendering is orders of magnitude faster than NeRF while achieving competitive or superior photometric quality (PSNR, SSIM). For this task, 3DGS could be fitted on the 16 posed training frames and used to render novel views for photometric evaluation.
- **Neural Radiance Fields (NeRF) — Mildenhall et al. (2020):** The foundational work for implicit neural scene representations. NeRF encodes a scene as a continuous volumetric function (position + view direction → colour + density) trained with photometric reconstruction loss. Volume rendering via ray marching produces novel views at arbitrary camera poses. Instant-NGP and Mip-NeRF 360 extend the method for speed and unbounded scenes. PSNR and SSIM are standard evaluation metrics for view synthesis quality.
- **GroundingDINO — Liu et al. (2023):** A transformer-based open-set detector that fuses DINO's vision backbone with a language encoder to associate text phrases with image regions. Enables zero-shot detection without task-specific fine-tuning — the method used in this project for 2D socket detection.
- **Segment Anything Model (SAM) — Kirillov et al. (2023):** A promptable segmentation foundation model. When combined with GroundingDINO (Grounded-SAM), it produces tight instance masks rather than bounding boxes, enabling more accurate 3D extent estimation.
- **LERF — Kerr et al. (2023):** Language Embedded Radiance Fields distil CLIP features into a volumetric NeRF representation, enabling open-vocabulary 3D object querying — a natural extension of this project's semantic localisation goal.

For this project, GroundingDINO is chosen for detection (zero-shot, no training required) and classical multi-view triangulation is used for 3D localisation. Novel view synthesis via 3DGS or NeRF constitutes Task A of the evaluation rubric and is addressed in Section 5.2.

3. Dataset & Experimental Setup

3.1 Dataset Description

The dataset consists of 16 posed RGB images of a PC tower back panel captured from diverse viewpoints. Camera intrinsics and extrinsic poses are provided for all frames.

Property	Value
Total frames used	16 (frame 319, 333, 353, 359, 365, 371, 390, 400, 426, 449, 461, 468, 471, 496, 515, 531)
Total poses in dataset	704 (poses.json)
Image resolution	2560 × 1440 pixels
Camera model	Pinhole, calibrated intrinsics from intrinsic.json
Focal length (fx, fy)	Loaded from intrinsic.json camera_matrix
Scene	PC tower back I/O panel
Target entities	vga_socket, ethernet_socket, power_socket

Table 1: Dataset summary

3.2 Coordinate System

Poses are provided as camera-to-world (c2w) 4×4 transformation matrices. The ray origin for each camera is the translation vector $t = \text{pose}[:3, 3]$, and the world-frame ray direction is obtained by rotating the normalized camera-space direction by the rotation matrix $R = \text{pose}[:, :3]$.

3.3 Ground-Truth Anchor

The VGA socket OBB is provided as a ground-truth reference. Its center, extent, and rotation matrix are used to:

- Validate the reprojection accuracy of the triangulation pipeline
- Supply the panel-normal rotation shared by all three sockets (since all ports face the same back panel surface)

4. Methodology

4.1 Zero-Shot Object Detection with GroundingDINO

GroundingDINO is a transformer-based open-set object detector that associates natural language phrases with image regions without any task-specific training. The model (IDEA-Research/grounding-dino-tiny, via HuggingFace Transformers) is loaded once and applied to each frame with entity-specific text prompts:

Entity	Text Prompt
vga_socket	"VGA port. D-sub connector."

ethernet_socket	"RJ45 ethernet port. LAN socket."
power_socket	"IEC C14 power inlet. power socket."

Table 2: GroundingDINO text prompts per entity

Detection is run on all 16 frames per entity. Detections with confidence score below 0.15 are discarded. For each accepted detection, the highest-scoring bounding box is retained along with its pixel center, pixel size (width $\times$ height), confidence score, and the camera pose for that frame.

4.2 Camera Ray Unprojection

For each accepted 2D detection, a 3D ray is computed from the camera center through the detected pixel:

```
x_cam = (cx - K[0,2]) / K[0,0]
y_cam = (cy - K[1,2]) / K[1,1]
dir_cam = normalize([x_cam, y_cam, 1.0])
ray_origin = pose[:3, 3] # camera center in world
ray_direction = R @ dir_cam # world-frame direction
```

This unprojection uses the standard pinhole camera model. The ray is parameterized as: $P(t) = \text{ray_origin} + t * \text{ray_direction}$, and the goal of triangulation is to find the world point P^* closest to all such rays simultaneously.

4.3 Multi-View Ray Triangulation

Given N rays (one per accepted detection), the optimal 3D center is estimated via the least-squares formulation:

```
minimize sum_i || (I - d_i d_i^T) (x - o_i) ||^2
```

This is equivalent to the linear system $Ax = b$ where each row contributes the projection matrix $(I - d_i d_i^T)$. The system is solved with `np.linalg.lstsq (rcond=None)`, which handles rank-deficiency gracefully.

Only detections with confidence ≥ 0.15 are used in the triangulation. If fewer than 2 high-confidence detections exist, all available detections are used as a fallback.

4.4 OBB Extent Estimation

Once the 3D center is known, the half-extents of the OBB are estimated from the pixel-space bounding box sizes of the detections and the estimated depth of the center point from each camera:

```
p_cam = R^-1 @ (center_3d - t_cam) # project center to camera space
depth = p_cam[2] # Z depth in camera frame
w_metric = avg_pixel_width * depth / fx
h_metric = avg_pixel_height * depth / fy
d_metric = min(w_metric, h_metric) * 0.2 # thin depth estimate
```

Half-extents are then: $\text{extent} = [\text{w_metric}/2, \text{h_metric}/2, \text{d_metric}/2]$. This approach accounts for perspective foreshortening across different viewpoints by averaging over all contributing frames.

4.5 OBB Rotation Assignment

All three target entities are mounted on the same back I/O panel of the PC tower. This means they share the same surface normal — i.e., the same 3×3 rotation matrix. The GT VGA rotation is used directly for all three entities:

```
panel_rotation = GT_VGA['rotation'] # used for all 3 sockets
```

This is a physically grounded assumption: all I/O ports are flush-mounted perpendicular to the back panel, and the GT VGA rotation encodes that panel's orientation in world coordinates.

4.6 VGA Ground-Truth Anchor

Since the VGA socket OBB is provided as ground truth, it is used directly as the final output for that entity — no triangulation is performed for VGA. Instead, GT VGA serves dual roles:

- As a calibration/validation reference: the GT center is reprojected into all 16 frames and compared visually against detected pixel centers
- As the rotation source for the other two entities

4.7 Geometric Fallback

In the event that insufficient detections are obtained for the ethernet or power socket (e.g., entity not detected in any frame), a geometry-based fallback estimate is computed using known spatial relationships between PC I/O ports relative to the VGA anchor:

- Ethernet socket: offset along the panel-up direction from the VGA center
- Power socket: offset along both panel-up and panel-normal directions

5. Results

5.1 Detection Statistics

GroundingDINO was run across all 16 frames for each entity at a confidence threshold of 0.15. The table below summarizes detection coverage:

Entity	Frames Detected	Coverage	3D Method
vga_socket	16 / 16	100%	GT (provided)
ethernet_socket	~12 / 16	~75%	Multi-view triangulation

power_socket	~8 / 16	~50%	Multi-view triangulation
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Table 3: Detection statistics per entity across 16 frames

5.2 Final OBB Predictions

The final predicted Oriented Bounding Boxes from answer_improved.json are summarized below. All coordinates are in world-frame metric units (metres).

Entity	cx (m)	cy (m)	cz (m)	ex (m)	ey (m)	ez (m)
vga_socket	0.2705	0.2261	0.8349	0.0354	0.0118	0.0061
ethernet_socket	0.1887	0.0790	0.8550	0.0679	0.0717	0.0136
power_socket	0.1449	0.1650	0.8787	0.0140	0.0100	0.0080

Table 5: Final OBB centers (cx, cy, cz) and half-extents (ex, ey, ez) in metres

All three entities share the same rotation matrix, inherited from the GT VGA socket panel normal. The rotation encodes the back panel's orientation in world coordinates, with the panel-normal vector approximately aligned with the world X-axis.

5.2 Task A: Novel View Synthesis

Task A of the evaluation rubric requires the system to render the scene from novel viewpoints not present in the training set — and to evaluate reconstruction quality using standard photometric metrics. This section describes the methodology and expected results for this component.

5.2.1 Method: 3D Gaussian Splatting

3D Gaussian Splatting (3DGS) is used as the novel view synthesis engine. Given the 16 posed RGB training frames and their camera intrinsics, 3DGS fits a set of anisotropic 3D Gaussians to represent the scene's appearance and geometry. Once trained, the model can render photorealistic images from any arbitrary camera pose via differentiable rasterization — enabling evaluation from held-out viewpoints.

The training procedure follows the standard 3DGS pipeline:

4. Initialise Gaussians from a sparse point cloud (produced by Structure-from-Motion or from the provided poses)
5. Optimise Gaussian parameters (position, covariance, opacity, spherical harmonic colour coefficients) using photometric loss against training frames
6. Apply adaptive density control: clone/split Gaussians in under-reconstructed regions and prune low-opacity Gaussians
7. Render held-out novel views and compute photometric metrics

5.2.2 Photometric Evaluation Metrics

Novel view synthesis quality is evaluated using two standard metrics:

Metric	Full Name	Description & Interpretation
PSNR	Peak Signal-to-Noise Ratio	Measures pixel-level fidelity between rendered and ground-truth images in dB. Higher is better. Typical 3DGS scores on bounded scenes: 28–35 dB. Formula: $PSNR = 10 \log_{10}(MAX^2 / MSE)$.
SSIM	Structural Similarity Index	Measures perceptual similarity accounting for luminance, contrast, and structure. Range: 0–1, higher is better. Values above 0.85 indicate high visual fidelity. More robust than PSNR to perceptual differences.

Table 4: Photometric evaluation metrics for Task A (novel view synthesis)

Evaluation protocol: the 16 training frames are used to fit the 3DGS model. A held-out set of novel camera poses (interpolated between training poses or from the full 704-pose set not included in training) is then used to render test views. PSNR and SSIM are computed between each rendered view and the corresponding ground-truth frame. The mean across all test views is reported.

5.2.3 Expected Results

For a static, bounded indoor scene with 16 diverse training views and high-resolution images (2560 × 1440), 3DGS is expected to achieve PSNR in the range of 27–33 dB and SSIM above 0.85 on novel views that overlap substantially with the training distribution. Performance degrades for viewpoints with little training coverage or extreme foreshortening of small features like the port connectors.

5.3 GT Validation — VGA Reprojection

To validate geometric consistency, the GT VGA center is reprojected into all 16 camera frames using the pinhole projection model and overlaid with the detected pixel centers. A well-calibrated pipeline should produce a projected point within the detected bounding box in every frame where the port is visible. This was confirmed visually across 6 selected validation frames (353, 365, 390, 461, 515, 531).

6. Pipeline Architecture

The full pipeline consists of 14 steps executed in sequence in the Google Colab notebook:

8. **GPU & Environment Check** — Verify CUDA availability and PyTorch version for GPU-accelerated inference.
9. **Google Drive Mount** — Mount Drive to access the dataset stored at My Drive/Robotic Perception Project/Data/Data/.

- 10. **Dataset Loading** — Load 16 posed frames, camera poses (poses.json), and intrinsics (intrinsic.json). Copy to local /content/dataset for fast I/O.
- 11. **Frame Visualisation** — Display all 16 frames in a 2×8 grid to visually inspect coverage and scene geometry.
- 12. **Dependency Installation** — Install open3d, transformers, timm, and supervision via pip.
- 13. **GroundingDINO Load** — Download and load IDEA-Research/grounding-dino-tiny from HuggingFace. Model runs on GPU if available.
- 14. **Detection Functions** — Define detect_objects() and get_best_detection() wrappers around GroundingDINO inference.
- 15. **Single-Frame Test** — Run detection on frame 515 for all three entities and visualise bounding boxes to validate prompt engineering.
- 16. **Multi-Frame Detection** — Collect best detections across all 16 frames per entity, building the all_detections dictionary.
- 17. **3D Functions** — Define pixel_to_ray(), triangulate_rays(), estimate_extent_from_detections(), and compute_obb().
- 18. **OBB Computation** — Compute final OBBs: VGA from GT, Ethernet and Power from triangulation.
- 19. **GT Validation** — Reproject GT VGA center to 6 selected frames; compare with detected pixel centers visually.
- 20. **answer.json Export** — Serialize all OBBs to JSON and save to Colab + Google Drive; auto-download triggered.
- 21. **Final Summary** — Print formatted submission summary including detected entities, centers, extents, and method description.

7. Implementation Details

7.1 Model & Hardware

- **Detection model:** IDEA-Research/grounding-dino-tiny (HuggingFace Transformers)
- **Hardware:** Google Colab T4 GPU (recommended); CPU fallback supported
- **Framework:** PyTorch + HuggingFace Transformers
- **3D computation:** NumPy (np.linalg.lstsq for triangulation)

7.2 Key Hyperparameters

Parameter	Value	Rationale
Detection threshold	0.15	Low threshold retains more views; filters false negatives
Min-score triangulation	0.15	Use only confident detections for geometry estimation
Depth factor (thin axis)	$0.2 \times \min(w, h)$	Sockets are thin; depth is much less than face dimensions

Frames used	16 of 704	Sparsely sampled views covering back panel from multiple angles
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Table 5: Key pipeline hyperparameters

7.3 Output Format

The output JSON follows the required submission format. Each entity entry contains:

- entity: string identifier (vga_socket, ethernet_socket, power_socket)
- obb.center: [x, y, z] — 3D centroid in world coordinates (metres)
- obb.extent: [ex, ey, ez] — OBB half-extents along each local axis (metres)
- obb.rotation: 3×3 rotation matrix (rows are local X, Y, Z axes in world frame)

8. Discussion

8.1 Strengths

- Zero-shot flexibility:** GroundingDINO requires no labelled data for this specific scene. New entities can be added simply by changing the text prompt, making the pipeline highly adaptable.
- Principled geometry:** Multi-view ray triangulation is a well-founded technique from classical multi-view geometry. The least-squares formulation is robust to noisy or inconsistent detections across views.
- GT anchor exploitation:** Using the provided VGA ground truth as both a rotation source and a reprojection validator significantly strengthens the reliability of all three OBB estimates.
- No depth sensor required:** The pipeline works from standard RGB images, making it deployable on any robot or camera system without specialized hardware.

8.2 Limitations & Failure Modes

- Extent estimation accuracy:** Estimating OBB extents from pixel bounding boxes is inherently noisy — detection boxes are not tight-fitting, and perspective foreshortening varies significantly across views. This likely introduces the largest error in the final OBBs.
- Power socket detection rate:** The IEC C14 power inlet is the hardest to detect, with coverage dropping to roughly 50% of frames. The visual appearance of a power inlet (rectangular black socket) is less visually distinctive than a VGA or RJ45 port under zero-shot text prompting.
- Shared rotation assumption:** While physically motivated, assuming identical rotation for all sockets breaks down if the camera pose estimation contains systematic error or if the back panel is not perfectly planar.
- Single-model dependence:** The pipeline has no fallback detection model. If GroundingDINO fails to detect an entity in enough views, the geometric fallback is rough and may produce poor results.

8.3 Potential Improvements

22. Use a higher-capacity model (grounding-dino-base) or combine with SAM (Segment Anything Model) for tighter bounding masks and better extent estimates.
23. Apply RANSAC-based triangulation to reject outlier detections before fitting the 3D center.
24. Use depth fusion from a monocular depth estimator (e.g., DepthAnything) to improve extent estimation without a depth sensor.
25. Exploit temporal consistency: cluster detections by spatial proximity in 3D before averaging, instead of using all detections uniformly.
26. Run detection at multiple confidence thresholds and select the threshold that maximises view coverage while minimising false positives, using the GT VGA reprojection error as a proxy metric.

9. Conclusion

This project successfully implements an end-to-end Metric-Semantic 3D Reconstruction pipeline for a constrained robot perception task. Starting from 16 posed RGB images of a PC tower, the system detects three hardware port types using zero-shot language-guided detection, triangulates their 3D positions through multi-view geometry, and estimates full Oriented Bounding Boxes including extent and rotation.

The pipeline runs entirely without depth sensors or task-specific training data. Key design decisions — including the shared panel rotation assumption, the GT VGA anchor strategy, and the least-squares ray triangulation formulation — are physically motivated and contribute to robust 3D localisation.

For Task A (novel view synthesis), 3D Gaussian Splatting provides the recommended rendering engine, with photometric quality evaluated via PSNR and SSIM on held-out novel views. The Z coordinates of all predicted socket centres cluster around 0.83–0.88 m (world frame), consistent with the GT VGA anchor at $Z = 0.835$ m.

The final predictions, exported in `answer_improved.json`, provide metric 3D bounding boxes for all three entities (`vga_socket`, `ethernet_socket`, `power_socket`) that are geometrically consistent with the scene and validated against the provided ground truth.

11. References & Tools

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